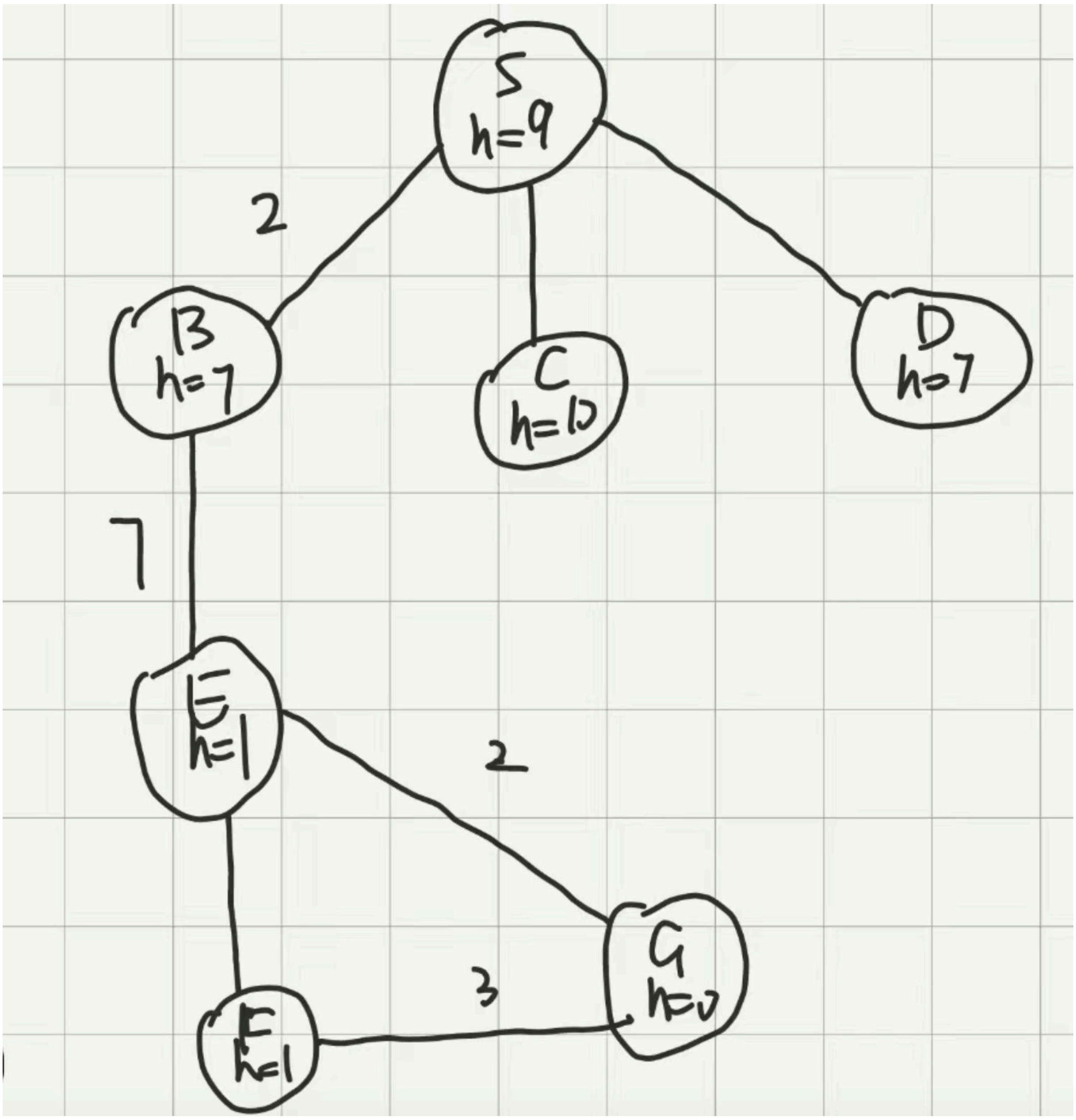
Artificial Intelligence 2024Spring – Final Exam

SuperbRain

2024-06-26

Problem 1: Search

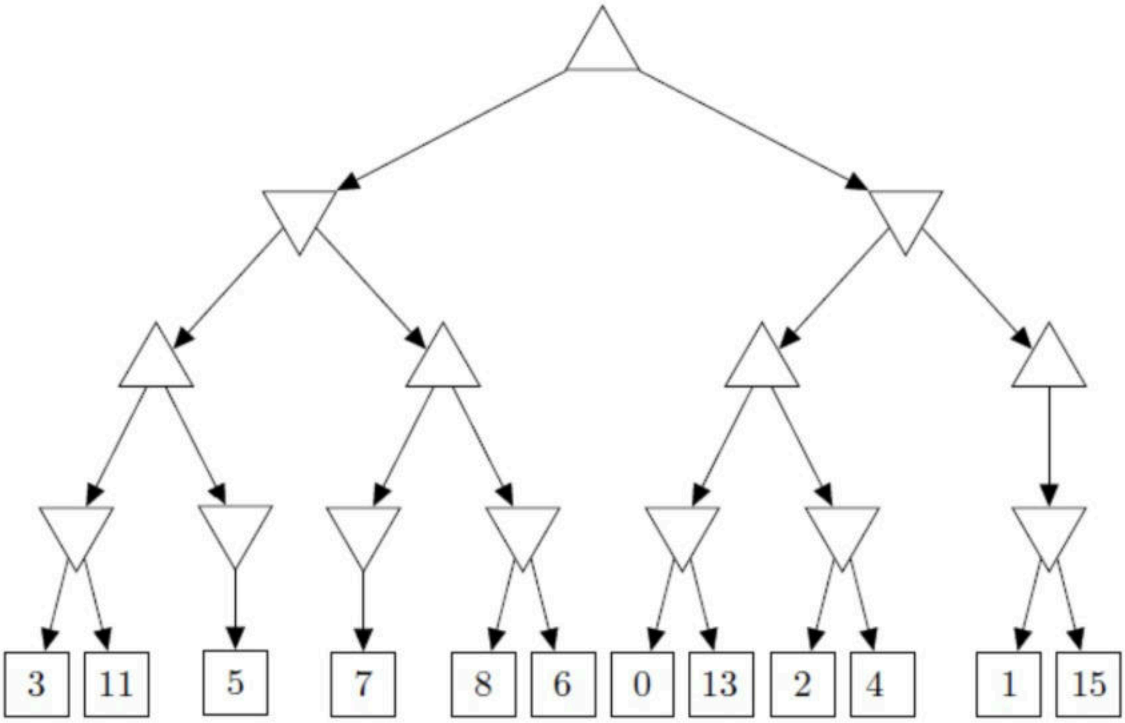


(1) Uniform-cost search

(2) Best first search

(3) A^* search

Problem 2: Minimax Search



(Numbers change to 11, 3, 5, 7, 8, 15, 0, 13, 2, 4, 1, 6)

alpha-beta pruning

Problem 3: CSP

$$X = \{X_1, X_2, X_3\}$$
$$D(X_1) = \{1, 2, 3\}, D(X_2) = \{1, 2, 3\}, D(X_3) = \{2, 3\}$$
$$C_1 : X_1 \neq X_2, C_2 : X_2 < X_3, C_3 : X_1 \neq X_3$$

(1) Draw the constraint graph for the given CSP;

(2) $X_1 = 2$, forward checking;

(3) $X_1 = 2$, arc consistency;

(4) Given a solution for this CSP or states that none exists.

Problem 4: Knowledge and Reasoning

$A \land \neg I, W \Rightarrow I, \neg A, \neg W \land \neg I \Rightarrow F$, Prove F.

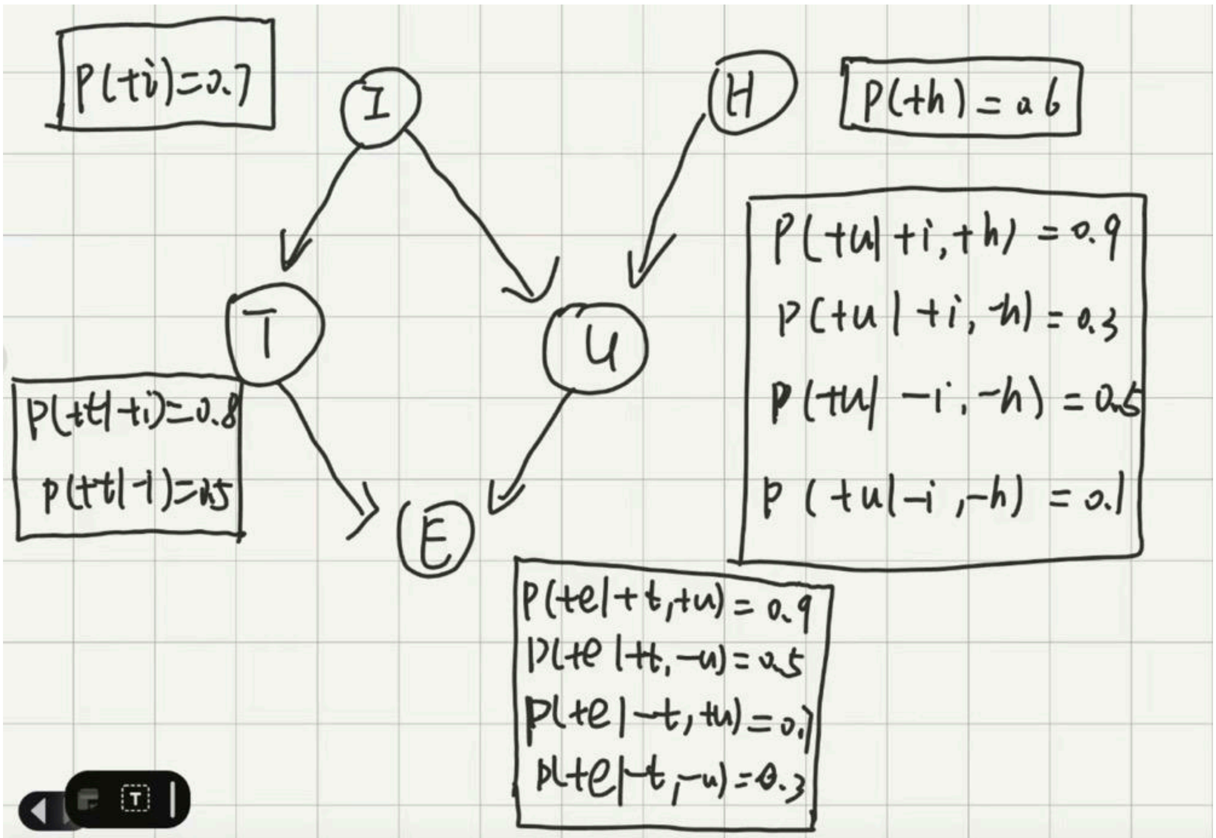
Problem 5: Decision Tree

Age	Temperature	Headache	Diagnosis
20-30	High	Yes	Positive
40-50	High	No	Positive
30-40	Normal	No	Negative
40-50	Low	No	Negative
20-30	Normal	Yes	Positive
50-60	High	No	Positive
40-50	Normal	No	Negative
20-40	Low	Yes	Negative

(1) Build a decision tree (do not need to show any computation and do not need to be optimal).

(2) Is this tree optimal? Explain in less than two sentences. If not, draw an optimal tree.

Problem 6: Bayesian Network



(1) $P(+u|+e)$, using elimination.

(2) Judging the independence of $(E, H|U)$, $(T, U|E, H)$, $(I, H|E)$, $(I, E|U)$.

Problem 7: Learning

7.1 ANN

$$(w_0, x_0, w_1, x_1, w_2, x_2) = (-3, 1, 2, -1, 4, 2)$$
$$f(\boldsymbol{x}) = \sigma(\boldsymbol{w}^T \boldsymbol{x})$$

Compute $\dfrac{\partial f}{\partial w_0}, \dfrac{\partial f}{\partial w_1}, \dfrac{\partial f}{\partial w_2}$.

7.2 CNN

7.2.a

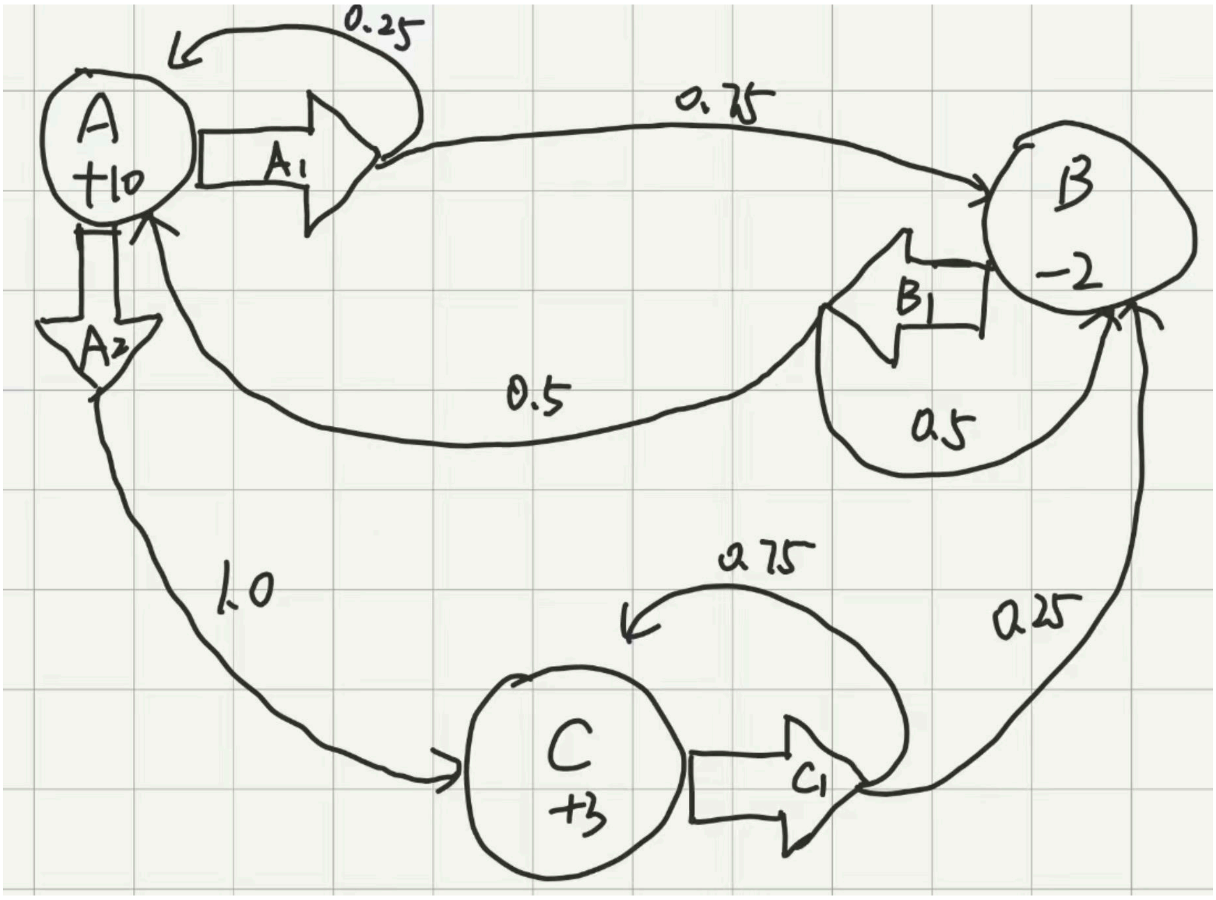
$$\text{Input} = \begin{pmatrix} 0 & 1 & 2 & 1 & -1 \\ 3 & 5 & 1 & 3 & 4 \\ 2 & 4 & 2 & 1 & 2 \\ 3 & 1 & 4 & 5 & 0 \\ 6 & 7 & -2 & 2 & 1 \end{pmatrix}, \text{Filter} = \begin{pmatrix} -1 & 0 & -1 \\ -2 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

stride=2, no padding, compute the output.

7.2.b

max-pooling, compute the output of Input.

Problem 8: MDP



$$U_1(s) = R(s)$$

$$U_{i+1}(s) = R(s) + \gamma \max_a \sum_{s'} T(s,a,s')U_i(s)$$

With $\gamma = 0.9$, compute:

(1) $U_2(A), U_2(B), U_2(C)$;

(2) $U_3(A), U_3(B), U_3(C)$